

Churn overview: 74% churned, repeat rate below 3%

Goal: quantify the revenue leak from churn · Model: XGBoost churn model (AUC 0.868) · Definition: customer snapshot — churned = no purchase for 180+ days

Total Customers

93K

Churned Customers

69K

Actual Churn Rate

74.1%

Predicted Churn Rate

80.9%

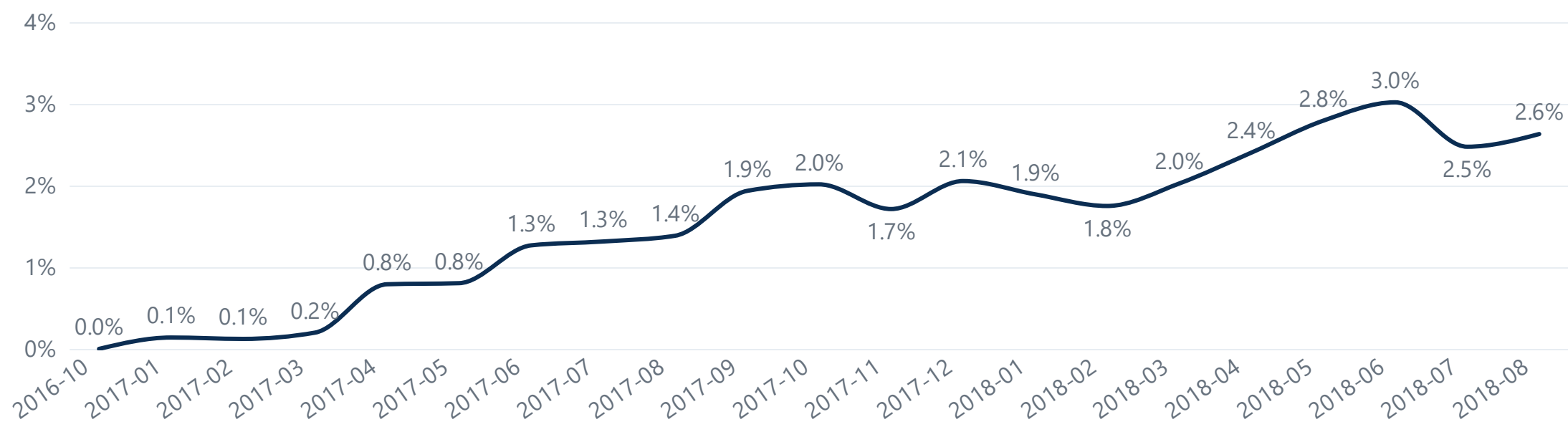
High Risk Customers

40K

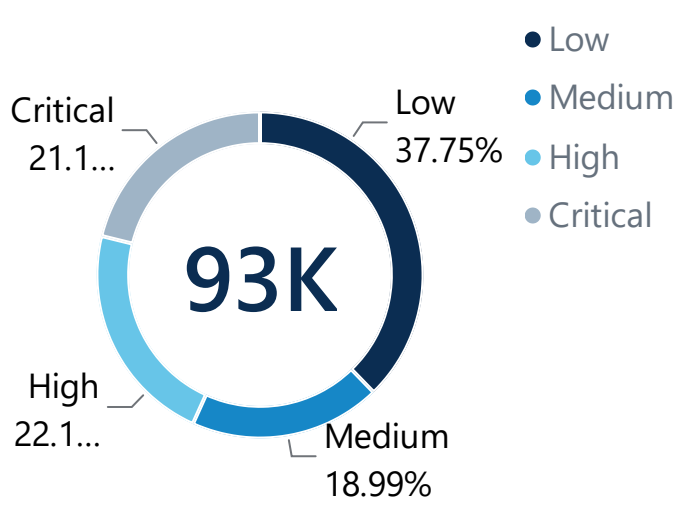
Winback Target GMV

622K

Repeat rate stays below 3% for 21 months



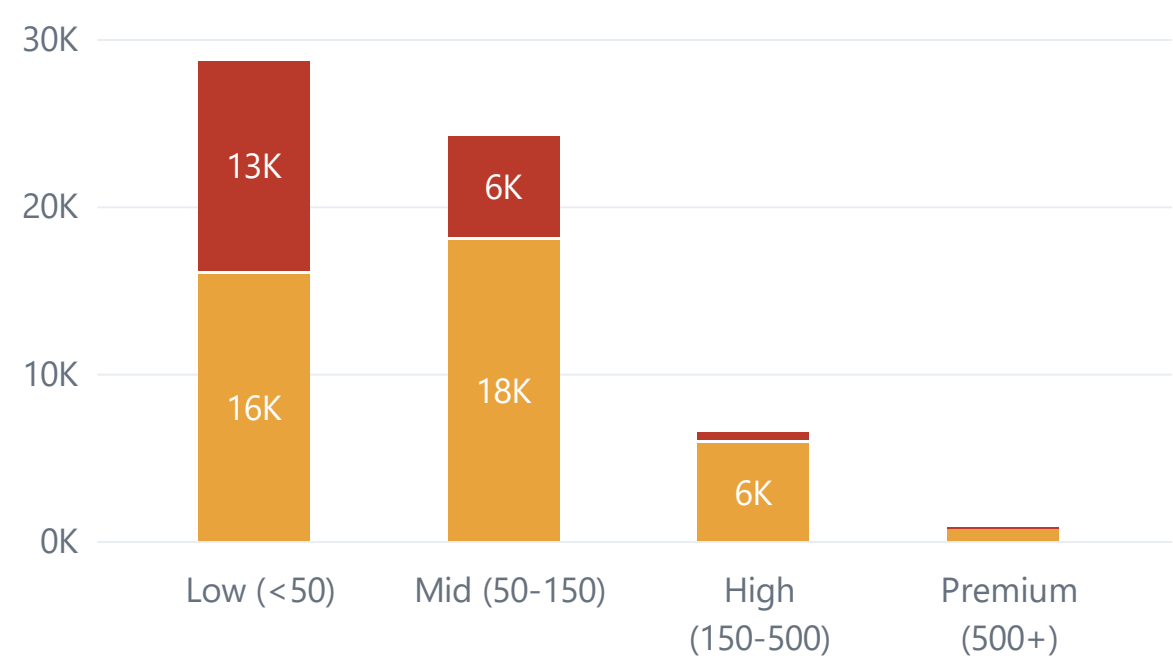
43% of customers sit in high or critical risk



97% ordered once; repeat buyers churn less

Frequency_Tie	Total_Customer	Churn
1 order	90,592	
2 orders	2,541	
3+ orders	225	
Total	93,358	

Risk is spread evenly across value tiers



Test set: 92.5% recall, 84.4% precision, AUC 0.868

Model_AUC	Model_Precision	Model_Recall	Model_F1
0.868	84.4%	92.5%	0.883

Month-1 retention is 0.5% and never recovers

Cohort_Mont	0	1	2	3	4	5
2017-01	100.00%	0.28%	0.28%	0.14%	0.42%	0.14%
2017-02	100.00%	0.18%	0.31%	0.12%	0.43%	0.12%
2017-03	100.00%	0.44%	0.36%	0.40%	0.36%	0.16%
2017-04	100.00%	0.62%	0.22%	0.18%	0.27%	0.27%
2017-05	100.00%	0.46%	0.46%	0.29%	0.29%	0.32%
2017-06	100.00%	0.49%	0.40%	0.43%	0.30%	0.40%
Total	100.00%	0.48%	0.34%	0.25%	0.29%	0.23%

74.1% of customers have churned (40,384 at high or critical risk) and the repeat rate has stayed below 3% for 21 months — the revenue base must be re-earned every

Why 180 days: 74% churn at 180, 94% at 90, 100% at 30

Caliber sensitivity over 93,358 customers, from the same snapshot that labels the model and the dashboard

Churn at 180 days · main caliber

74.1%

Churn at 90 days

93.6%

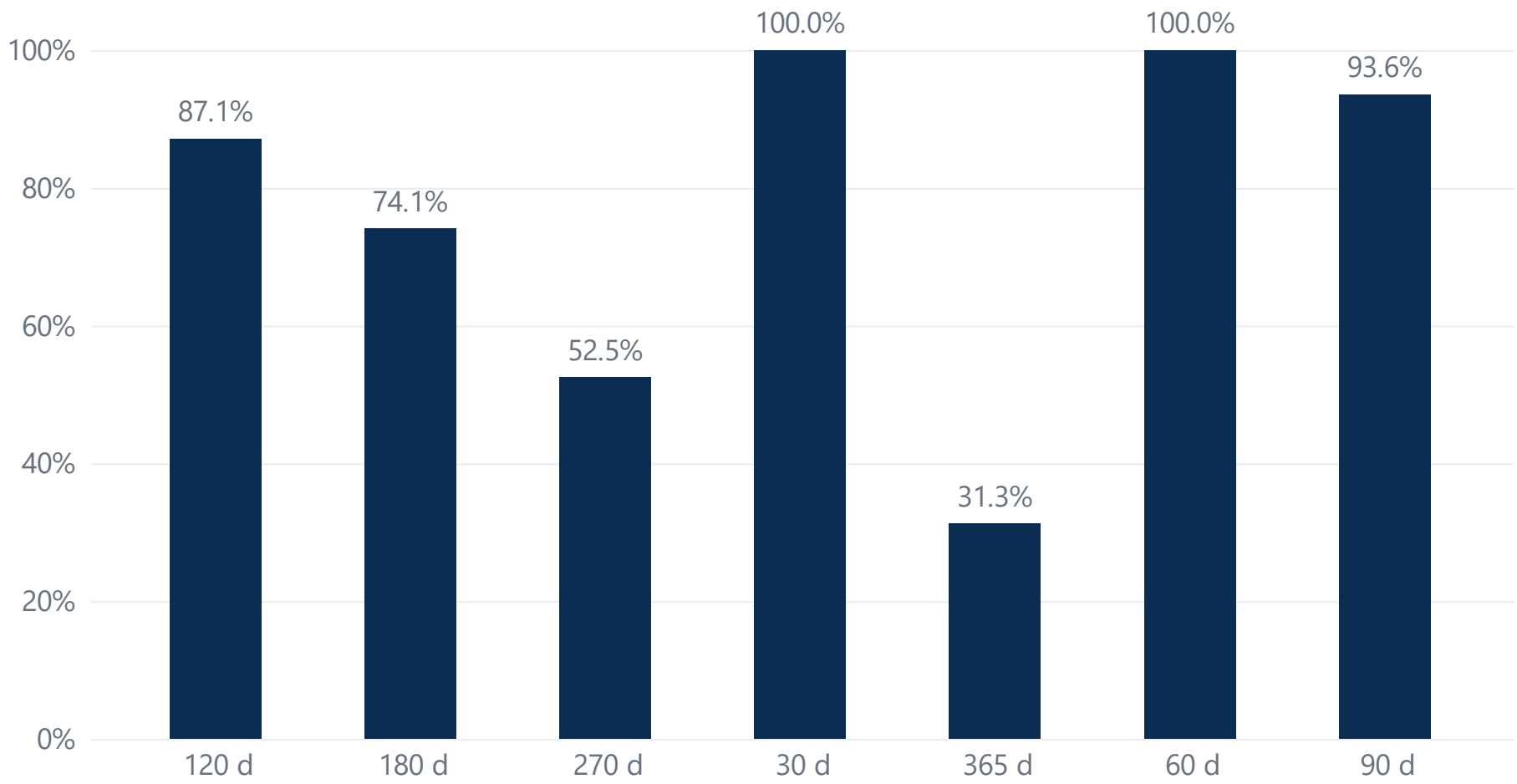
Retention at 180 days · healthy 30%

25.9%

Shortest recency in the snapshot · days

63

Churn runs from 100% at 30 days to 31% at 365; 180 days is the usable rung



Two things the reader should carry

Same-day multi-order customers: 803 of the 2,766 repeat customers (29.0%) placed every order on one day — a split cart, not a returning customer. The true cross-period repeat rate is 2.10% rather than 2.96%, and the value multiple moves 2.02x → 1.98x.

30 days still matters as an alerting scale, not a churn definition: only 29.6% of first-to-second purchases happen inside 30 days, so day 30 is when the customer is worth contacting

Three calibers answer three different questions

caliber	definition	value_text	use_case
Dormant (main)	Last purchase more than 180 days ago	74.1%	Churn rate, model label, cau
First-gap interval	First purchase to second purchase	74 days	Designing the intervention v
Repurchase	No second order within 180 days of the	97.4%	Whether a repurchase habit

Why 180 days

- 1 · 30 / 60 days cannot work here: every customer's last purchase is already more than 60 days old (shortest 63), so a 30-day window labels all 93,358 customers churned.
- 2 · 90 days is too tight: 93.6% churn is nearly everyone, so it adds no discrimination.
- 3 · 180 days matches the real repurchase interval: first-to-second purchase medians 74 days and p75 179 days (n = 2,066) — it covers three quarters of observed repurchases.
- 4 · 180 days is the benchmark-comparable rung: healthy retention is published at 30/90/180/365 days, and at 365 the observation window truncates the rate (68.7% against a 25% line, on a snapshot whose longest recency is 776 days).

Who churns: geography 7pp, delivery experience 20pp

Note: the dataset has no gender/age fields — interpretable behavioural tiers (value, frequency, satisfaction, delivery experience, freight) replace demographics

High Value Customers

24K

High Value Churned

18K

Churned Revenue Share

73.6%

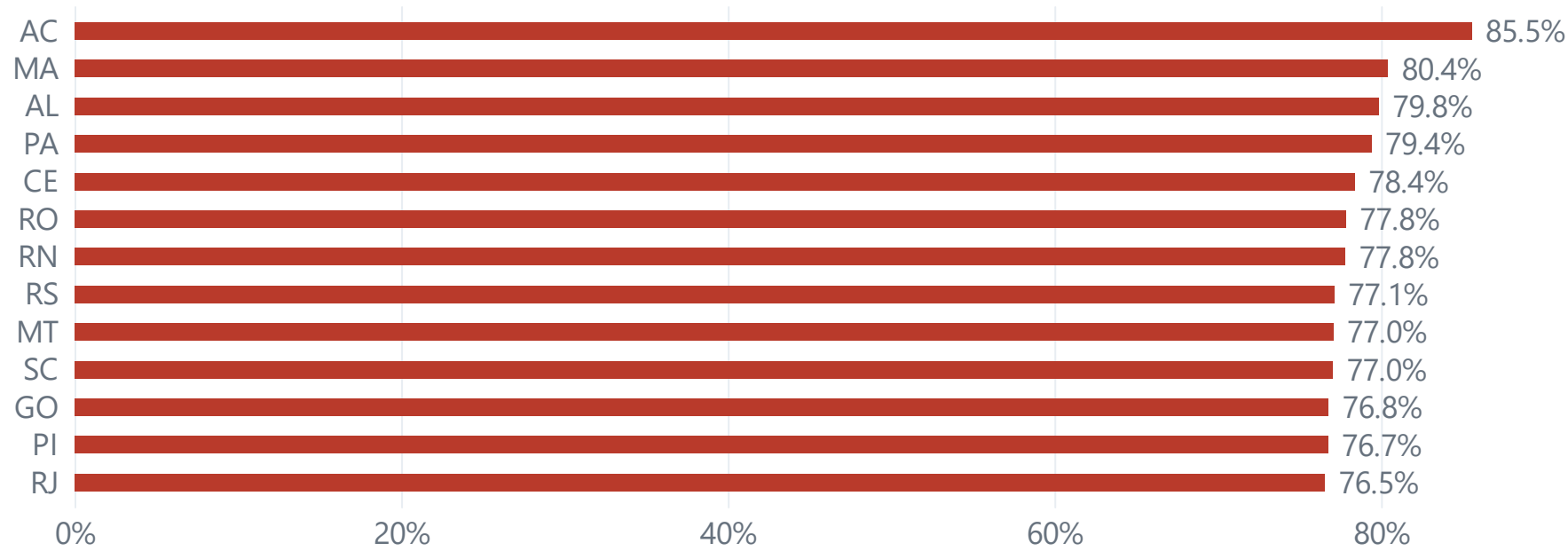
Logistics Driven Share

81.3%

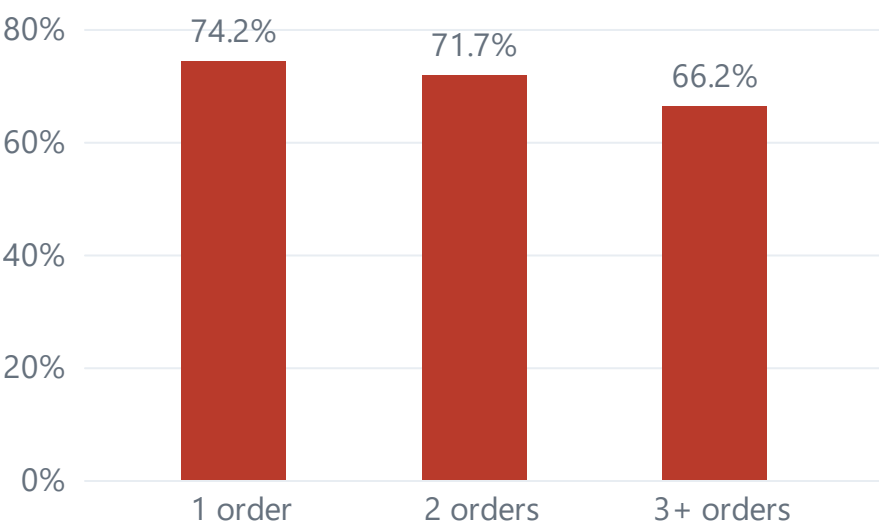
Geography Driven Customers

5K

MA, AL, PA and RO churn most; SP least



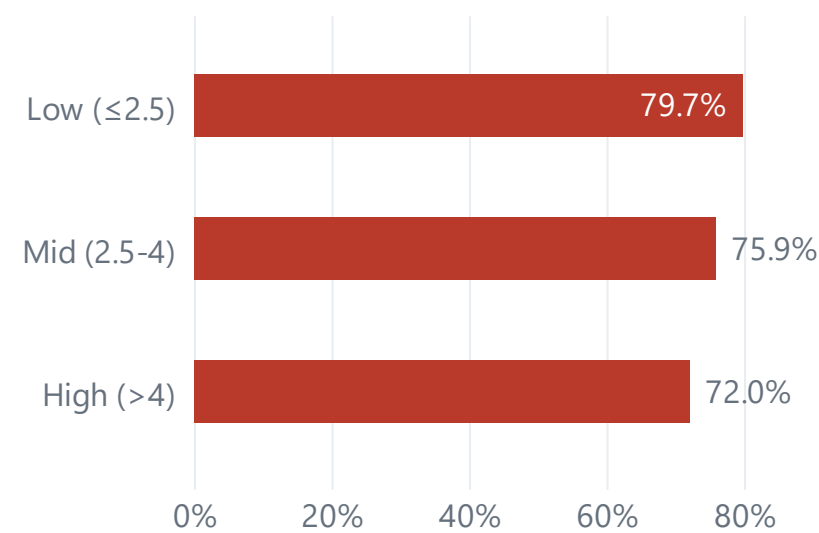
Purchase frequency barely moves churn (66% at 3+)



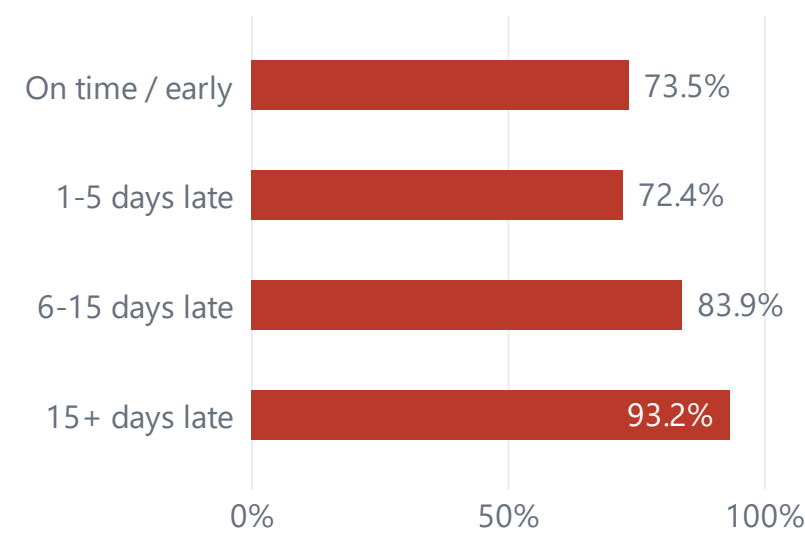
High-risk share is similar across value tiers

Value_Tier	Low	Medium	High	Critical
Low (<50)	10,050	1,294	3,158	12,768
Mid (50-150)	15,913	8,480	11,700	6,238
High (150-500)	7,808	6,487	5,116	684
Premium	1,471	1,471	656	64
Total	35,242	17,732	20,630	19,754

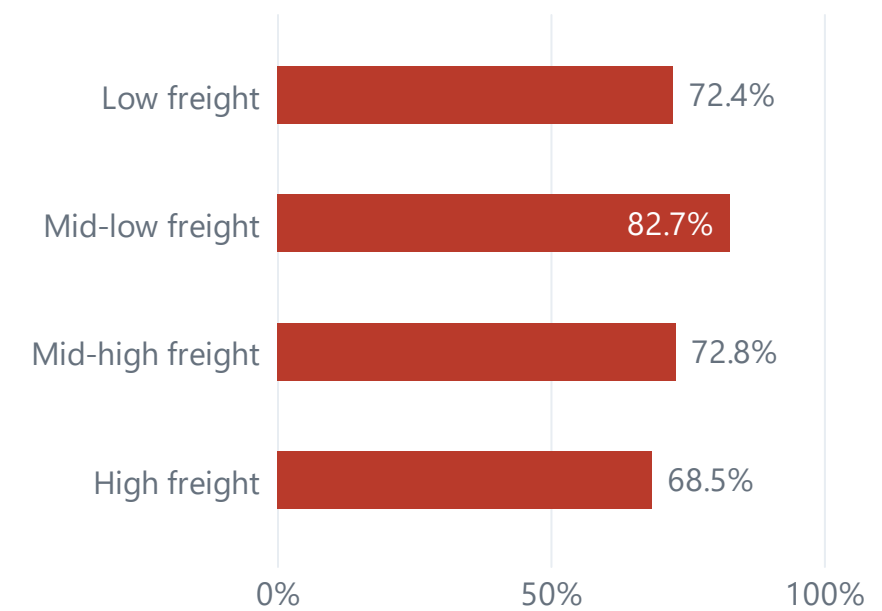
Low-satisfaction customers churn 8pp more



Churn reaches 93% when delivery is 15+ days late



Freight tier barely moves churn



Retained and churned customers look alike — the difference is when they stopped

Actual_Outcome	Total_Customer	Total_Revenue
Churned	69,162	9,768,642
Retained	24,196	3,507,570
Total	93,358	13,276,212

Geography moves churn by 7pp and delivery experience by 20pp, while value tier barely matters — interventions should target experience, not basket size.

Why they churn: freight and delay carry 59% of attribution

TreeSHAP attribution for every customer — the top 3 factors (TOP3 features) drive differentiated win-back actions

Model AUC

0.868

Model Recall

92.5%

Logistics Driven Customers

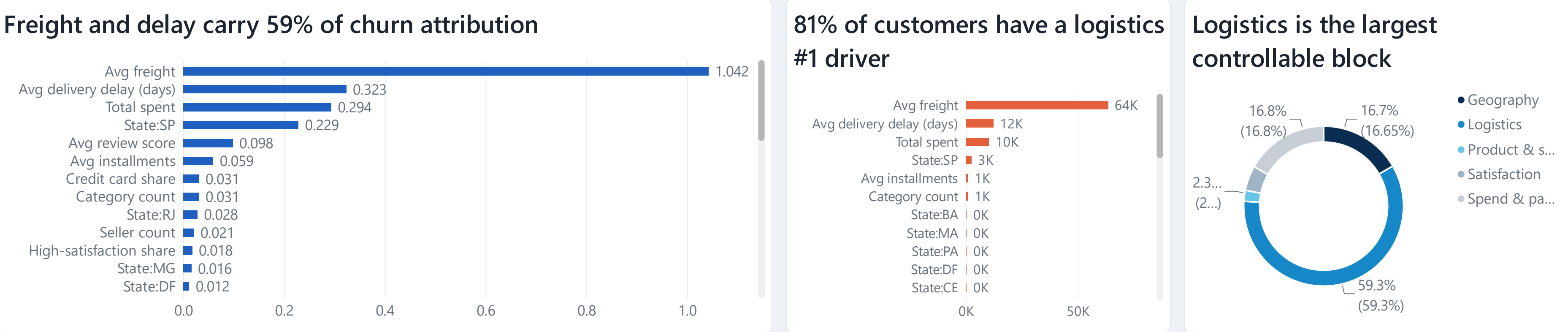
76K

Logistics Driven Share

81.3%

Satisfaction Driven Customers

158



Churned: +22% freight and +7.8 days delay

Actual_Outcome	Overall_Avg_Freight	Overall_Avg_Delay	Overall_Avg_R...
Churned	19.69	-10.9	
Retained	21.59	-11.9	
Total	20.18	-11.1	

Every customer carries TOP3 factors and direction

Customer_State	Churn_Probability	Risk_Tier	Top1_Factor	Top2_Factor	Top3_Factor	Total_Spent
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	12.90
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	13.65
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Category count	20.30
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	10.99
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	State:SP	19.36
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	11.55
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	12.25
RJ	99.9%	Critical	Avg freight	Avg delivery delay (days)	Total spent	3.54

Freight and delay carry 59% of the model's attribution and 81.3% of customers have a logistics #1 driver; the model identifies churners with 92.5% recall.

Satisfaction breaks at 3-7 days late: low scores 26% to 62%

Five delay buckets over 99,992 orders, plus the slowest categories and states

low-score share, 3-7 days late

61.6%

low-score share, 7+ days late

78.3%

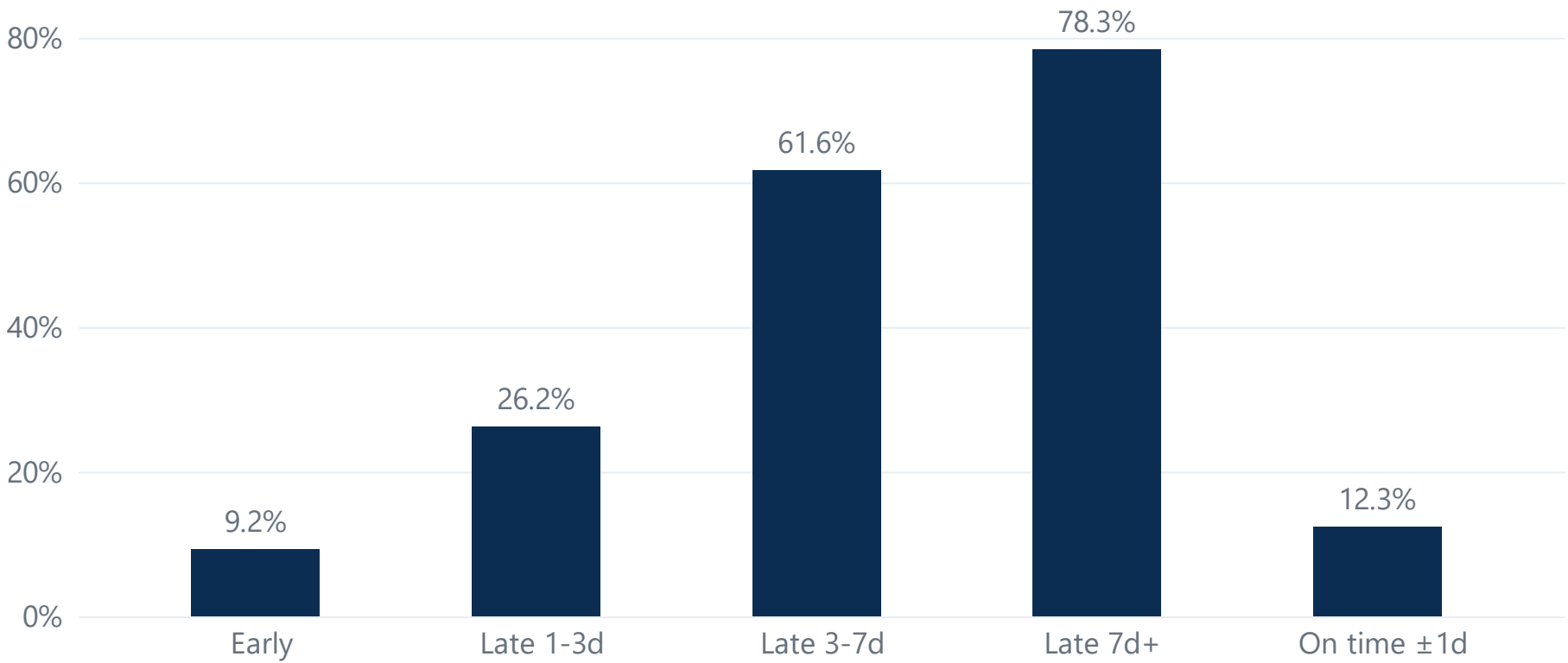
slowest state, average days

26.3

slowest category, average days

20.8

Low scores step from 26% to 62% between 3 and 7 days late



Office furniture is slowest at 20.8 days and 25.4% low scores

group_name	n_orders	avg_days	avg_review	low_pct
	1,533	12.7	3.94	19.4
audio	359	13.3	3.83	21.7
bebes	2,967	12.4	4.08	14.9
cama_mesa_banho	10,985	12.7	3.92	18.3
casa_conforto	430	13.5	3.85	19.8
casa_construcao	502	12.2	3.96	17.4

91.9% of orders arrive early; the late tail is where scores die

bucket_label	order_count	share	avg_review	low_pct
Early	88,552	91.90	4.294	9.2
Late 1-3d	1,335	1.39	3.493	26.2
Late 3-7d	1,790	1.86	2.305	61.6
Late 7d+	3,245	3.37	1.731	78.3
On time ±1d	1,431	1.49	4.035	12.3

AM is slowest at 26.3 days against a 12-day platform norm

group_name	n_orders	avg_days	avg_review	low_pct
AL	398	24.4	3.84	21.4
AM	145	26.3	4.22	13.8
BA	3,246	19.2	3.93	17.0
CE	1,276	21.2	3.94	17.3
DF	2,089	13.0	4.13	13.2
ES	1,978	15.6	4.08	13.8
GO	1,963	15.5	4.10	13.2
MA	716	21.4	3.84	19.8
MG	11,354	12.0	4.19	11.7
MS	710	15.6	4.17	13.1
MT	882	17.9	4.15	12.7

Voice of the customer: 60% of complaints are logistics

Source: low-score reviews (rating ≤ 2); pain point, sub-issue and improvement suggestion extracted by an LLM

Negative Reviews

300

Severe Dissatisfaction Reviews

12

Severe Dissatisfaction Share

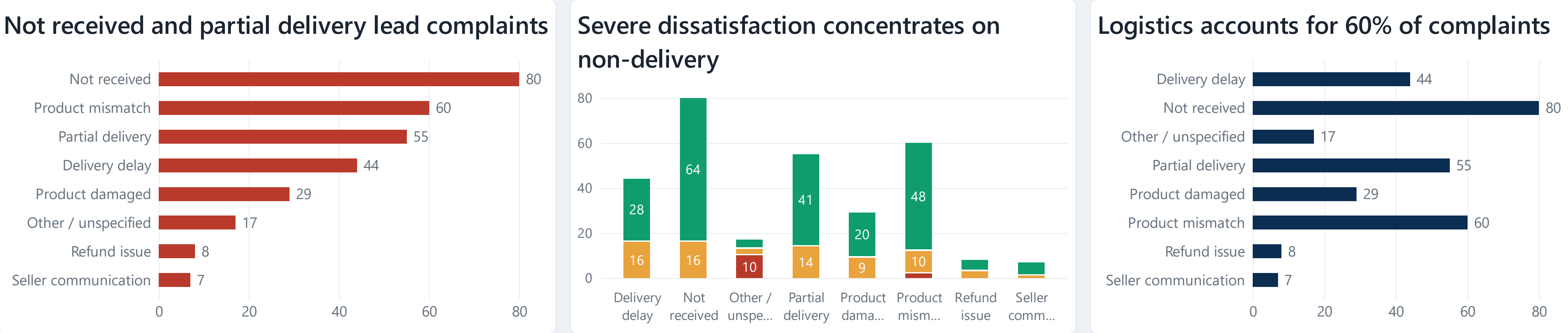
4.0%

Logistics Pain Reviews

179

Avg Review Rating

1.2



Every complaint carries an actionable fix

Pain_Point	Sub_Issue	Dissatisfaction_Level	Review_Keyword	Review_Translation
Delivery delay	approved_then_cancelled	Mild	POR DEMORA CANCELAMENTO	Order approved 10/06, cancelled 10/17 for delay, but then they ship it. Cant get contact via sta
Delivery delay	at_post_office_wants_refund	Mild	produto esta nos correios... quero meu estorno	Paid shipping, product stuck at post office far from home. Wants refund.
Delivery delay	business_days_overdue	Mild	mais de 20 dias úteis	Ordered over 20 business days ago, still not delivered.
Delivery delay	chase_required	Dissatisfied	Tive que entrar em contato com a empresa pra	Had to chase the company to receive product. Wont buy this way again.
Delivery delay	correios_slow	Mild	dar 1 mês e o Correios não entregou	Regrets buying due to using Correios; almost 1 month and not delivered.
Delivery delay	extreme_delay	Mild	esperar mais de 60 dias	Never imagined waiting 60+ days; paying for a product that may never arrive is the same as n
Delivery delay	extreme_delay_caps	Mild	DEMOROU MUIIIITOOOOOOOOO	Took a REALLY REALLY long time to arrive (caps for emphasis).
Delivery delay	first_issue_loyal	Mild	primeira vez que estou insatisfeito	Long-time lannister customer, first issue with delivery time.

60% of complaints are logistics (not received, partial, delay) — the same ranking the model produced, and only 4% are severe, so most are fixable service failures.

What is identified, and what the ranking can and cannot do

DAG identification with refutation checks, Qini / AUUC on the T-Learner holdout (n = 8,997), and the decile check behind it

PSM ATT · late delivery → churn

0.0389

PSM ATT · late delivery → satisfact...

-0.464

Qini curve, end · baseline 0

-0.029

AUUC · target > 0.05

-0.0035

repeat events, treated · 20:1 versus...

12

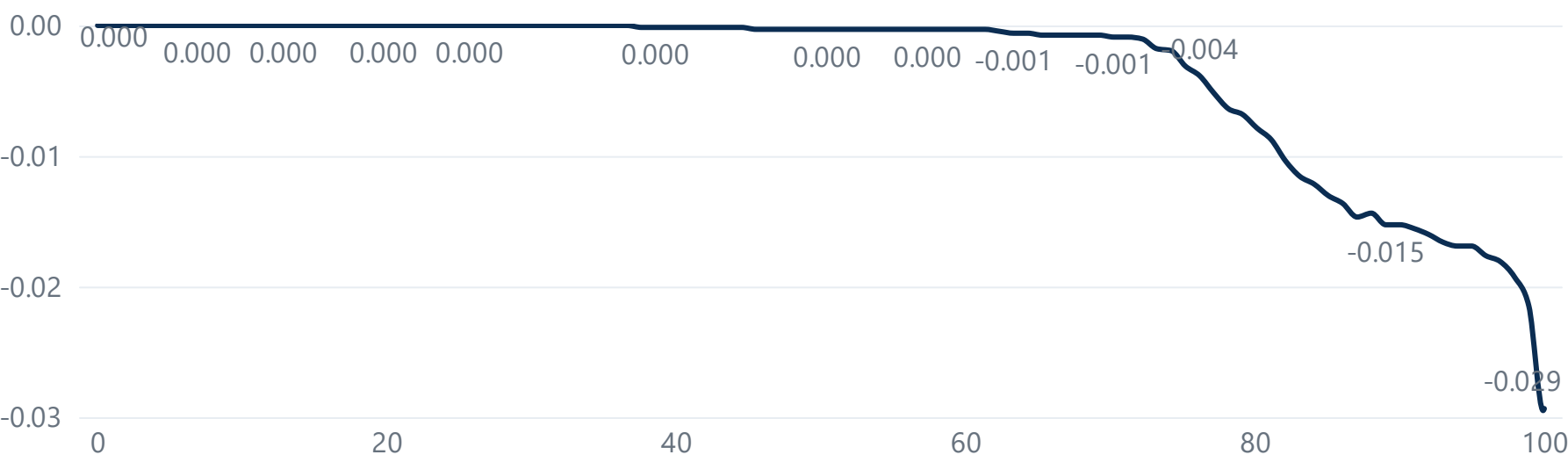
refutation checks passed

3

Only two edges are identified; the repeat link is not

from_node	to_node	edge_kind	effect_label
Avg delay days (instrument)	Late delivery (treatment)	instrument	
Churn (> 180 days)	GMV (not estimated)	not estimated	
Customer state	Late delivery (treatment)	confounding	
Customer state	Satisfaction (mediator)	confounding	
Late delivery (treatment)	Churn (> 180 days)	identified	+3.9pp
Late delivery (treatment)	Repeat purchase	estimated, not established	+0.11pp n.s.
Late delivery (treatment)	Satisfaction (mediator)	identified	-0.46 pts

Cumulative gain never rises above baseline



D1–D3 hold 2,222 of 2,249 treated customers and all 12 events

decile	pred_uplift	n_treated	events_treated	repeat_treated_pct	n_control	events_control	repeat_control_pct	band
1	-0.065	787	6	0.76%	113	113	100.00%	D1-D3
2	-0.011	834	5	0.60%	66	66	100.00%	D1-D3
3	0.061	601	1	0.17%	299	49	16.39%	D1-D3
4	0.119	14	0	0.00%	885	4	0.45%	D4-D10
5	0.132	1	0	0.00%	899	0	0.00%	D4-D10

What is identified, and what is not

layer	method	metric	estimate	unit	p_value	significant
Effect	DML (DoWhy)	Repeat purchase	-0.0006	probabilit		False

Why the ranking fails: sample size, not direction

The outcome is extremely sparse: repeat runs at 2.96% overall and only 12 of 2,249 treated customers repeat — a 20:1 event imbalance against 234 control events. The T-Learner's CATE is therefore unstable and Qini/AUUC fall below baseline: a precision problem, not an absent effect (PSM, DML and IV agree).

Above benchmark on every step but repeat: 3.85%

17 benchmark pairs, four customer classes, the model card and four estimator families side by side

repurchase vs the 30% norm

--

90-day retention vs 40%

-33.6

consecutive repeat buyers, monthly avg

0.31%

classes outside the healthy band

3

Every upstream node beats benchmark; repeat is 26.2pp short

layer_label	key_label	benchmark	olist	gap_pp
Annual churn	Annual churn vs e-commerce	25.0	74.1	49.1
Annual churn	Annual churn vs marketplace	40.0	74.1	34.1
Annual churn	Annual churn vs subscription	5.0	74.1	69.1
Customer class	Consecutive repeat (monthly avg)	30.0	0.3	-29.7
Customer class	Deep-wake (monthly avg)	0.0	0.5	0.5
Customer class	New buyers (monthly avg)	30.0	98.1	68.1
Customer class	Short-recall (monthly avg)	15.0	1.1	-13.9
Funnel node	Approved	99.0	99.8	0.8
Funnel node	Delivered	95.0	98.8	3.8
Funnel node	Not delivered	99.0	99.8	0.8

Satisfaction spreads 66-90% by state but repeat stays 3-5%

group	n_start	N6_step	N7_step
automotivo	3,900	79.6	2.82
BA	3,397	72.8	3.68
beleza_saude	8,850	81.1	3.58
brinquedos	3,873	81.1	3.24
cama_mesa_banho	9,415	74.3	5.93
DF	2,160	78.8	3.65
ES	2,043	77.2	4.06
esporte_lazer	7,738	81.6	5.00

Three of four classes sit outside the healthy band

label	avg_pct	benchmark_min	benchmark_max	deviation_pct	is_healthy
Consecutive repeat	0.31	30.0	100.0	-99.5	False
Deep-wake	0.53	0.0	5.0	-78.8	True
New buyers	98.07	30.0	40.0	180.2	False
Short-recall	1.08	15.0	25.0	-94.6	False

Recency defines the label, so it is excluded from the features

item	value
AUC / F1	0.868 / 0.882
Class weight	scale_pos_weight = 0.35
Features	38 (leakage-free)
Leakage control	recency dropped (it defines the label)
Precision / recall	0.844 / 0.925
Split	70 0017 / 22 319

Churn and satisfaction effects replicate; the repeat link is zero everywhere

method	question	estimate	significance
DoWhy doubly robust (DML)	Late → repurchase	-0.0006	≈0
DoWhy linear regression	Late → repurchase	-0.0034	≈0

Actions & watchlist: ROI 0.95x–4.34x, and only at the upper bound

Strategy ROI = annual incremental GMV under a stated repeat elasticity. The repeat link is not identified (IV p = 0.61), so the band runs 0.17x–4.34x; the uplift ranking is reported with its Qini/AUUC metrics, not shipped as a

High Risk Customers

40K

Critical Risk Customers

20K

High Risk Revenue

4M

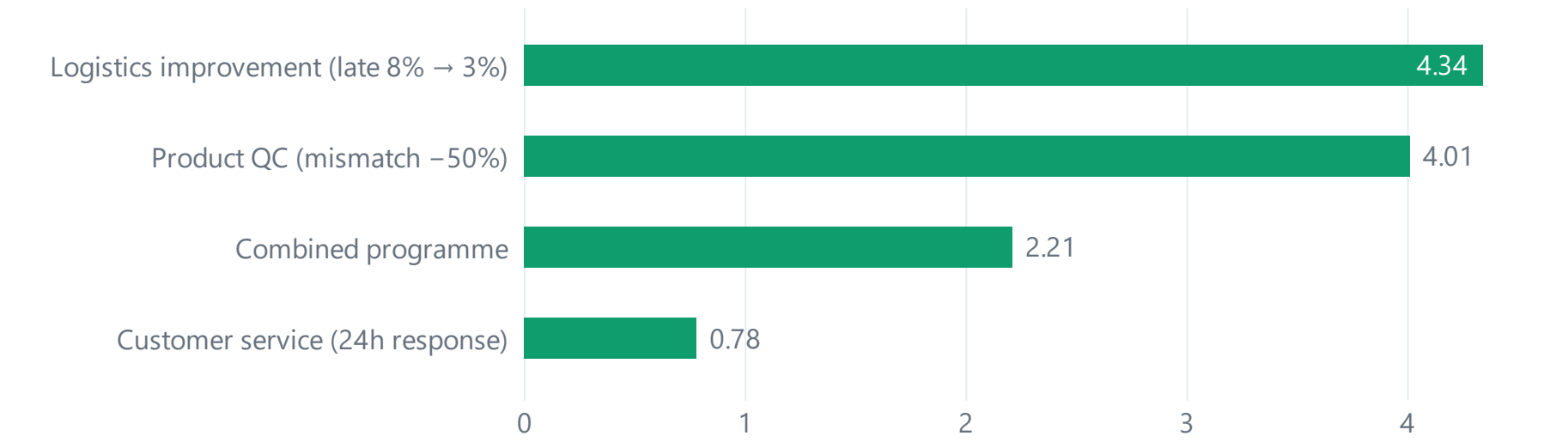
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622K

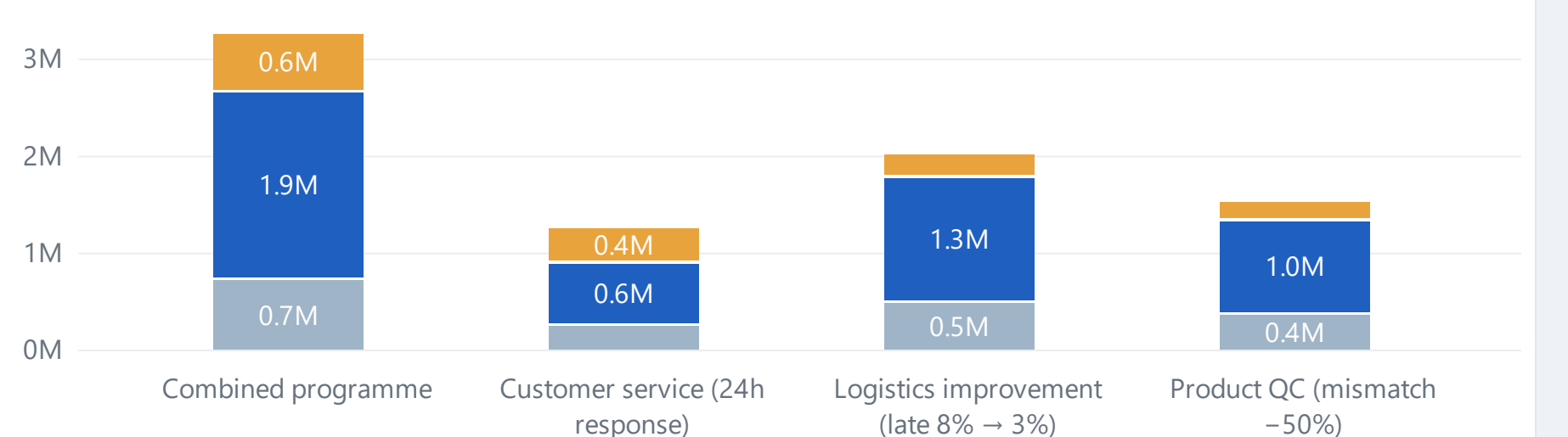
Blended ROI

3.45

Logistics 4.34x ROI; product QC 4.01x



Programme ROI as a 1.46–4.00 pp elasticity band



Watchlist sorted by churn probability, ready to act

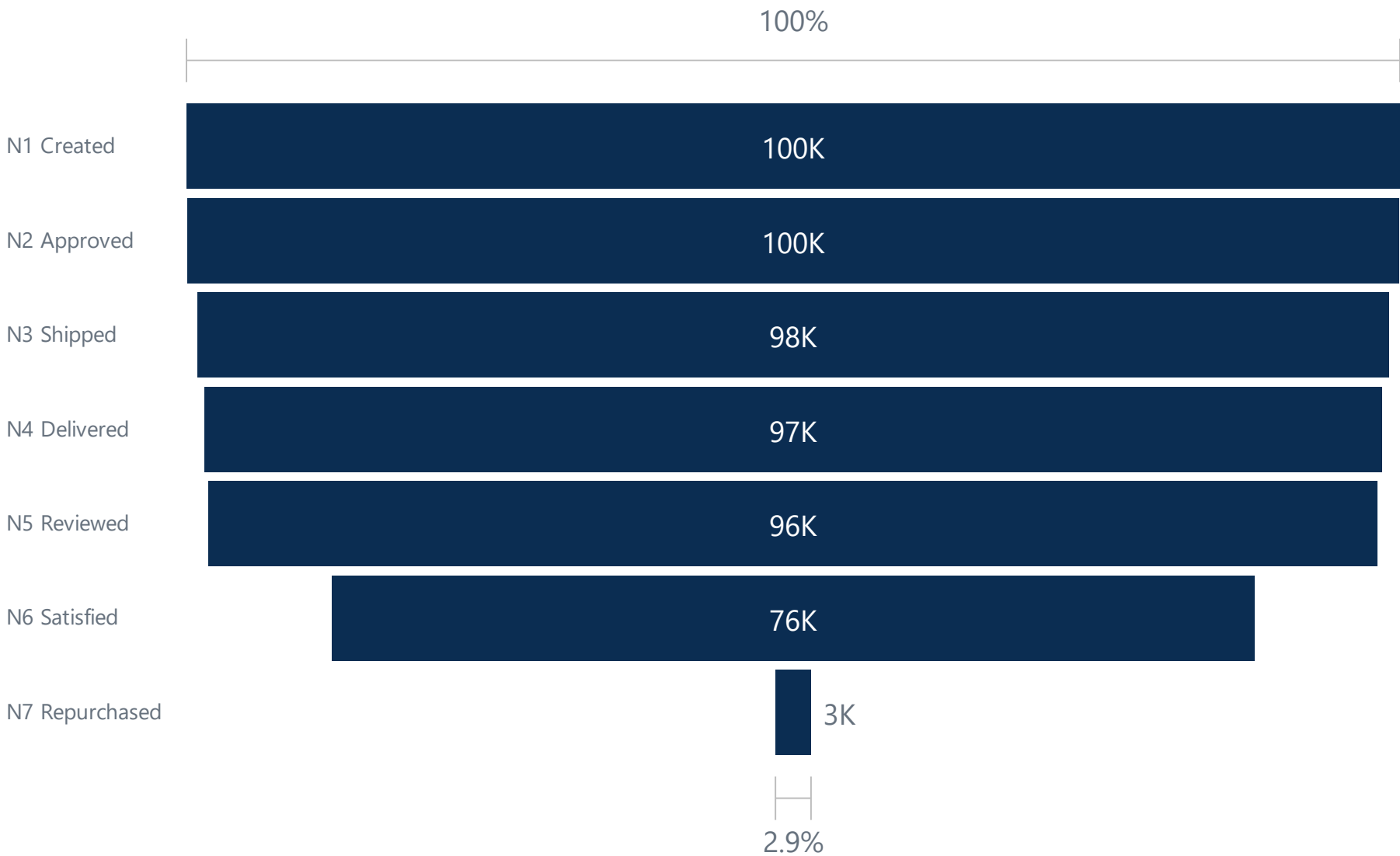
Customer_ID	Customer_State	Churn_Probability	Risk_Tier	Total_Spent	Order_Count	Avg_Review_Scor	Top3_Factor_Combo	Recommended_Action
2403ebb49b70a5690aa8fb7d1df202db	RJ	99.9%	Critical	12.90	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
651da97b2f7575d76efe82f970e57116	RJ	99.9%	Critical	13.65	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
ed5b0d9aa093cc966424056ec8c10c4c	RJ	99.9%	Critical	20.30	1	1.00	Avg freight / Avg delivery delay (days) / Category	Urgent: exclusive coupon
49399a471271928d84eb38ad823c8dd	RJ	99.9%	Critical	10.99	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
02ce856522310e96cddf763e086f3221	RJ	99.9%	Critical	19.36	1	1.00	Avg freight / Avg delivery delay (days) / State:SP	Urgent: exclusive coupon
2313a05d4b08ad58b737b4d9df881419	RJ	99.9%	Critical	11.55	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
6dfd53d0b64aa10fd66b8a7a4ad4461c	RJ	99.9%	Critical	12.25	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
84b4ea17e1ba0e30dd2b6e71937b682	RJ	99.9%	Critical	3.54	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
cc086e1370c1b8891f3df560c359ad62	RJ	99.9%	Critical	13.65	1	1.00	Avg freight / Avg delivery delay (days) / Total spent	Urgent: exclusive coupon
6edf0b82cff21956cb04a4502f093224	RJ	99.9%	Critical	18.90	1	2.00	Avg freight / Avg delivery delay (days) / State:SP	Urgent: exclusive coupon

At the fitted elasticity none of the four clears its cost; logistics and product QC reach payback only at the upper bound (0.95x–4.34x). Start with logistics and product

Seven funnel nodes show where volume survives and where it does not

Snapshot 2016-09 → 2018-10 · 99,992 orders · step and cumulative conversion per node

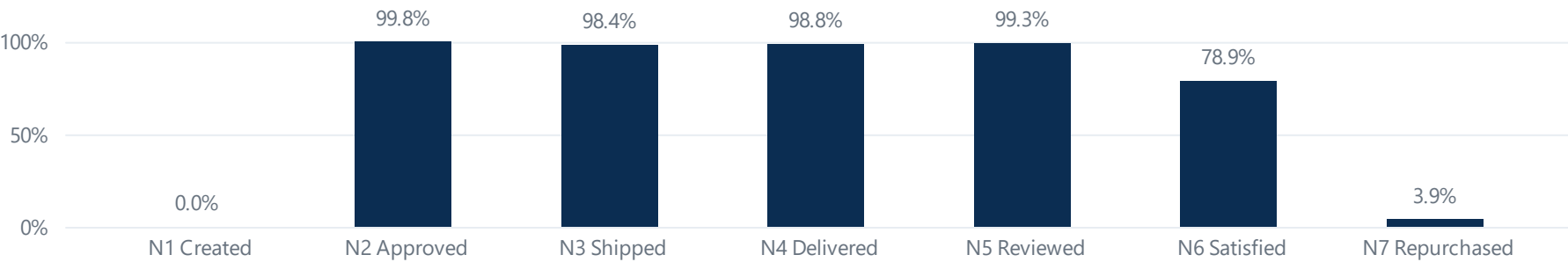
Satisfaction and repurchase are the only two real drops



Volume survives to review; satisfaction costs 21%, repurchase 96%

node_label	stage_count	step_conversion_pct	cumulative_conversion_pct
N1 Created	99,992	0.0%	0.0%
N2 Approved	99,831	99.8%	99.8%
N3 Shipped	98,185	98.4%	98.2%
N4 Delivered	96,990	98.8%	97.0%
N5 Reviewed	96,338	99.3%	96.4%
N6 Satisfied	76,033	78.9%	76.0%
N7 Repurchased	2,930	3.9%	2.9%

Only satisfaction and repurchase are real drops



Returning customers split three ways: consecutive buying is only 0.46% of buyers

